5.12 By definition, torque is given by

$$\vec{N} = \int \vec{x} \times (\vec{J} \times \vec{B})\, d^3x = \int [\, \vec{J}(\vec{x}\cdot\vec{B}) - \vec{B}(\vec{x}\cdot\vec{J})\,]\, d^3x .$$

Here, $\vec{J}$ is the current density on the inner loop with radius b, and $\vec{B}$ is the magnetic induction from the outer loop with radius a on the inner loop. It is easy to see that $\vec{x}\cdot\vec{J} = 0$ on the inner loop. Then,

$$\vec{N} = \int \vec{J}(\vec{x}\cdot\vec{B})\, d^3x = b\int B_r \vec{J}\, d^3x ,$$

where, from Section 5.5, we know for a current loop on the x-y plane,

$$B_r = \frac{\mu_0 I a}{2r} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n n!} \frac{r_<^{2n+1}}{r_>^{2n+2}} P_{2n+1}(\cos\theta)$$

$$= \frac{\mu_0 I}{2} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n n!} \frac{b^{2n}}{a^{2n+1}} P_{2n+1}(\cos\theta) .$$

with $r_< = r = b$, and $r_> = a$. In the coordinate system, it is not very easy to determine θ for the inner loop, as it makes an angle with the x-y plane. We can rotate the whole system around the y-axis by an angle of α so that the inner loop now lies in the x-y plane. The original z-axis now points in the $(\theta, \phi) = (\alpha, 0)$ direction, and the points on the inner loop can be parameterized by $(\theta', \phi') = (0, \phi')$. Then, the angle γ between the original z-axis and any point on the inner loop becomes $\cos\gamma = \cos\alpha + \sin\alpha \cos\phi'$. By the summation formula, Eq. (3.68), we have

$$P_l(\cos\gamma) = P_l(\cos\theta) P_l(\cos\theta') + 2 \sum_{m=1}^{l} \frac{(l-m)!}{(l+m)!} P_l^m(\cos\theta) P_l^m(\cos\theta') \cos(m(\phi - \phi'))$$

$$= P_l(\cos\alpha) P_l(0) + 2 \sum_{m=1}^{l} \frac{(l-m)!}{(l+m)!} P_l^m(\cos\alpha) P_l^m(0) \cos(m\phi') .$$

Also, the current density on the inner loop in the new coordinate system is

$$\vec{J} = -J_\phi \sin\phi' \,\hat{i} + J_\phi \cos\phi' \,\hat{j} ,$$

where $J_\phi = I' \sin\theta' \,\delta(\cos\theta') \dfrac{\delta(r'-b)}{b}$. Then,

$$\vec{N} = b\int B_r \vec{J}\, d^3x = b\int r'^2 dr' \int_{-1}^{1} d(\cos\theta') \int_0^{2\pi} d\phi' \left(-J_\phi \sin\phi' \hat{i} + J_\phi \cos\phi' \hat{j}\right)$$

$$\times \frac{\mu_0 I}{2} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n n!} \frac{b^{2n}}{a^{2n+1}}$$

$$\times \left(P_{2n+1}(\cos\alpha) P_{2n+1}(0) + 2\sum_{m=0}^{2n+1} \frac{(2n+1-m)!}{(2n+1+m)!} P^m_{2n+1}(\cos\alpha) P^m_{2m+1}(0) \cos(m\phi') \right)$$

We can see that the integration only has non-zero component in $\hat{j}$-direction. Also, the ϕ'-integration is non-zero for $m=1$ only. Noticing $P_{2m+1}(0) \equiv 0$, we finally have

$$\vec{N} = \frac{\mu_0 I I' b}{2} b\pi \, \hat{j} \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)!!}{2^n n!} \frac{b^{2n}}{a^{2n+1}} \cdot 2 \frac{(2n)!}{(2n+2)!} P^1_{2n+1}(\cos\alpha) P^1_{2n+1}(0).$$

$$= \frac{\mu_0 \pi I I' b^2}{2a} \hat{j} \sum_{n=0}^{\infty} \left(\frac{b}{a}\right)^{2n} \frac{(-1)^n (2n+1)!!}{2^n n!} \frac{1}{(n+1)(2n+1)} P^1_{2n+1}(\cos\alpha) \frac{(-1)^{n+1} \Gamma(n+\frac{3}{2})}{\Gamma(n+1)\Gamma(\frac{3}{2})}$$

Since $\dfrac{\Gamma(n+3/2)}{\Gamma(n+1)\Gamma(\frac{3}{2})} = \dfrac{(2n+1)!!}{2^n (n+1)!}$, then we can write the above expression as

$$\vec{N} = -\hat{j} \frac{\mu_0 \pi I I' b^2}{2a} \sum_{n=0}^{\infty} \left(\frac{b}{a}\right)^{2n} \frac{n+1}{2n+1} \left[\frac{\Gamma(n+\frac{3}{2})}{\Gamma(n+2)\Gamma(\frac{3}{2})} \right]^2 P^1_{2n+1}(\cos\alpha).$$

Here we have assumed that the current on the inner loop is counter clockwise. There will be a sign flip if the direction of the current flips.